

Demonstration of Proposed Features

Date	2 july 2026
Team ID	XXXXXX
Project Name	Voice-Based concept Understanding Analyser
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status	Demonstrated	Remarks
1	User Authentication.	Secure user login and registration to access the application.	Implemented	Yes	Working Successfully
2	Audio Upload & Recording	Users can upload or record voice explanations directly through the interface.	Implemented	Yes	Working Successfully
3	Speech-to-Text Transcription	Converts speech into text using OpenAI Whisper.	Implemented	Yes	Accurate transcription achieved
4	Semantic Similarity Analysis	Compares user explanation with the	Implemented	Yes	AI-based concept evaluation

		reference concept using Sentence-BERT			
5	Audio Feature Analysis	Extracts pause ratio, RMS energy, speech rate, and filler words using Librosa.	Implemented	Yes	Speech quality metrics generated
6	Evaluation Dashboard & PDF Report	Displays transcript, scores, feedback, visualizations, and downloadable PDF report.	Implemented	Yes	Interactive dashboard with report generation

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%